

Which tool(s) should I select?

Based on a few questions you can make informed decisions which tools you can select to perform enrichment, pathway, and/or network analysis. This is just a guideline and does not guarantee that the decision is correct even if all answers were.

It is good practice to use 2 or 3 tools and compare the results.

KEGG refers to the API usage e.g. with clusterprofiler in R.

1. User expertise

1.1 Are you fluent in R

 MetaboAnalyst (R 4.0 or web 6.0), KEGG, FELLA, RaMP-DB, and any other tool

 MetaboAnalyst 6.0, RaMP-DB, MetScape, IDSL.GOA

1.2 Do you prefer a web-based tool

 MetaboAnalyst 6.0, RaMP-DB, IDSL.GOA

 MetaboAnalyst R 4.0, MetScape, KEGG, FELLA

1.3 Do you have knowledge in Cytoscape

 MetScape and any other tool

 All tools

2. Origin of data:

2.1 Do you work with data that has clinical relevance

 IDSL.GOA and any other tool

 all tools except IDSL.GOA

2.2 Which organism do you study?

Check if organism specific metabolite sets are available.

3. What kind of biological interpretation do I want?

3.1 Enrichment analysis

- Metabolite set enrichment
- MetaboAnalyst, Ramp-DB, IDSL.GOA

3.2 Pathways

- Canonical metabolic pathways (e.g. glycolysis, TCA)
- Easy to interpret and widely accepted
- MetaboAnalyst, KEGG, Metscape, RaMP-DB

3.3 Networks

- Relationships between metabolites, reactions, and enzymes
- Hypothesis-generating and exploratory
- Metscape FELLA and IDSL.GOA

3.4 Biological functions beyond pathways

- Ontology-based interpretation
- IDSL.GOA